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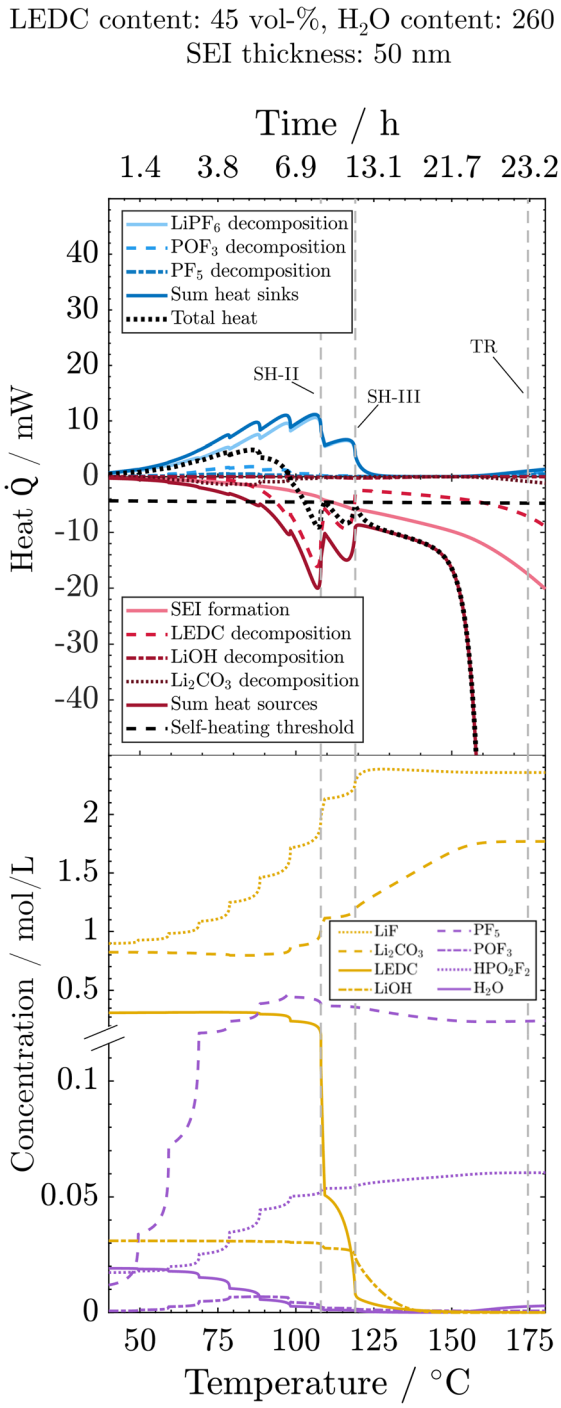

Fig. 4 Processes during the ARC test for the reference case: evolution of heat sinks (blue) and heat sources (red) and related concentrations, as well as the total heat (black) and the self-heating threshold (horizontal dashed line). Vertical dashed lines indicate events marked in Fig. 2. The y-axis break is at 0.12 mol m^{-3}.

directly consumed. In contrast, the LiF concentration increases until all $LiPF_6$ is decomposed and then stays constant. The SEI is thus completely inorganic at higher temperatures. Concentrations of HF, $LiPF_6$ and Li_2O are either 0 (HF) or can be deduced from LiF ($LiPF_6$) or LiOH (Li_2O) progressions. For the sake of clarity, they are shown in the ESI.†

Having understood the process interplay for the reference case, we now evaluate how their dependence on SEI composition and thickness, and water content can explain the observed change in thermal self-heating and thermal runaway behaviour.

As SEI composition impacts mainly the SEI formation and SEI decomposition reactions, their corresponding heat is presented in Fig. 5a. Li_2CO_3 decomposition is only marginally affected by changes in the SEI composition and is thus displayed together with other reactions in the ESI.†

A higher LEDC volume fraction leads to notably earlier on-set of LEDC decomposition. This leads to the earlier self-heating on-set at 98 °C (SH-I) for the 90 vol% LEDC case compared to the reference case (45 vol% LEDC). For 90% LEDC, heat production from this reaction decreases much slower after reaching self-heating because there is still a significant amount of LEDC in the SEI (see the ESI†). Together with the higher SEI formation reaction, this can explain why the 90 vol% LEDC case needs no further heating step to proceed to thermal runaway. The large fraction of LEDC in the initial SEI and its rapid but not complete consumption lead to a significant drop in SEI thickness to 60% before SEI formation sets in and rebuilds the SEI. The decline is much more significant than for the reference case, where *ca.* 80% of the SEI, mostly inorganic, remains and is subsequently rebuilt. For the inorganic case, *i.e.* 0 vol% LEDC in the initial SEI, the missing exothermic heat from the decomposition of the initial LEDC and the slow exothermic SEI formation lead to additional heating steps. At around 128 °C, marked by SH-IV, eventually, self-heating of the battery sets in due to higher exothermic SEI formation rates at this temperature. As no LEDC is present in the initial SEI in this case, and as Li_2CO_3 decomposition is negligible and compensated by its production, SEI thickness increases monotonously. In all three cases, during proceeding self-heating >120 °C, the heat from LEDC decomposition is smaller than that from SEI formation. It can be concluded that LEDC content in the SEI is strongly impacting thermal safety, as the self-heating onset is strongly impacted by LEDC decomposition and formation rates, and thus by LEDC availability.

We now discuss the impact of initial SEI thickness on heat evolution from LEDC decomposition and reformation (Fig. 6a) and on the resulting changes in SEI thickness (Fig. 6b). From Fig. 2b, we know that a thicker initial SEI leads to a higher self-heating temperature. This is counterintuitive as more SEI, *i.e.*, more LEDC, means more reactants for low-temperature decomposition. Indeed, more SEI leads to more LEDC decomposition (Fig. 6a); yet, during the seek period, LEDC decomposition heat decreases to similar values for all SEI thicknesses, whereas, for the thin SEI, exothermic SEI formation is almost double that of the thicker SEIs. The thickness-dependent SEI formation rate, eqn (3), is the key to explain why a thin SEI still leads to lower self-heating temperatures. The thinner the SEI is, the higher the formation rate and thus the heat produced from SEI formation reactions. SEI thickness can therefore be seen as a beneficial property of the SEI to prevent further formation and early thermal runaway. Our findings are also in good agreement with ARC tests of cycling-aged cells in the temperature range of 35–45 °C from the literature by Feng *et al.*,[23] Feinauer *et al.*,[22]

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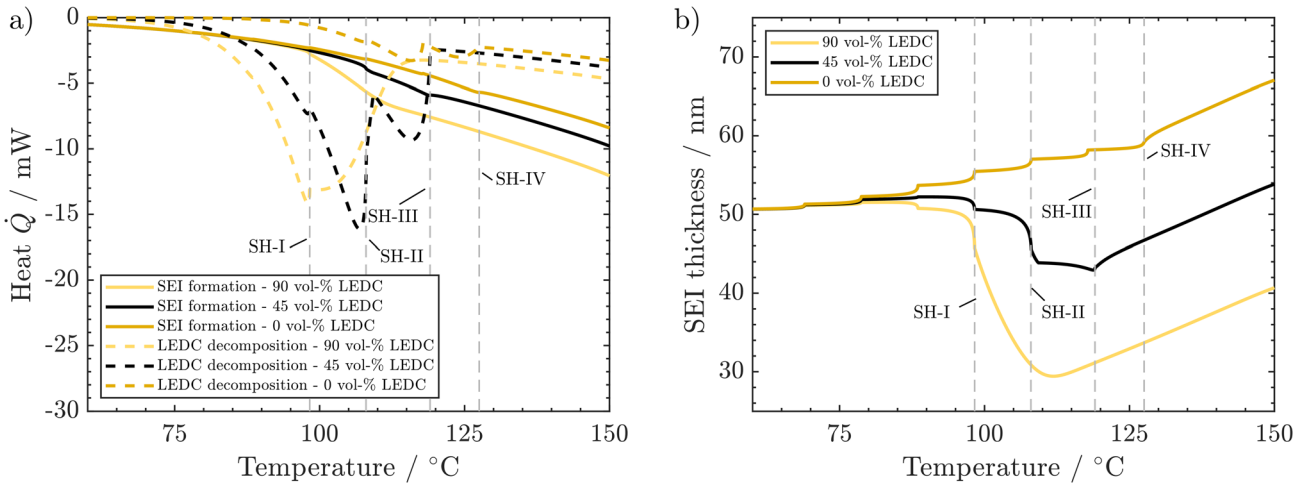

Fig. 5 Effect of SEI composition on behaviour during the ARC test: (a) produced heat from LEDC decomposition and SEI formation. (b) Corresponding changes in SEI thickness. Conditions: 260 ppm water, 50 nm thick SEI.

Börner *et al.*,[20] and Waldmann *et al.*[21] All studies independently found that these ageing procedures lead to an increase in the on-set temperature of self-heating by 15–20 °C when compared to their fresh reference. Our study now delivers an explanation for this increase: when cells age, the SEI becomes thicker and more inorganic; both effects have been shown here to lead to a delayed self-heating temperature. Röder *et al.*[56] in contrast found a lowered self-heating temperature for calendaric-aged cells at 60 °C. Our model may explain this behaviour also: either the SEI had a much more total amount of LEDC, probably dissolved also in the electrolyte, or the LEDC had reacted at 60 °C and left a thin, less inhibiting SEI.

Finally, we analyse the effect of H_2O content deeper to understand why low and medium water impurities lead to the same temperature evolution while high water content leads to earlier self-heating. PF_5, POF_3 and Li_2CO_3 decomposition reactions (PFD, POFD, and ISD) happen almost simultaneously. Their added values are endothermic and increase with water content (Fig. 7a). It can also be observed that the higher the water content, the earlier the on-set of these reactions. In the case of dry and reference H_2O concentrations, the released heats are almost identical and very small. From *ca.* 80 °C onwards, in the wet electrode case, most of the endothermic heat is still small and released in a temperature range where no exothermic counterpart exists. With increasing impurity concentrations, the temperature gradient after sustained self-heating changes from 4.4 °C h^{-1} (dry case) to 4.65 °C h^{-1} (medium dried case) to 4.1 °C h^{-1} (wet case). These changes are connected to the increased concentration of LiOH, which decomposes exothermically, and the corresponding decrease in self-heating temperature (wet case). First, when moving from the dry case to the medium dry case, the temperature gradient only increases slightly which can be explained only by a small increase in LiOH content from 0.4 vol% to 0.6 vol%. However, in

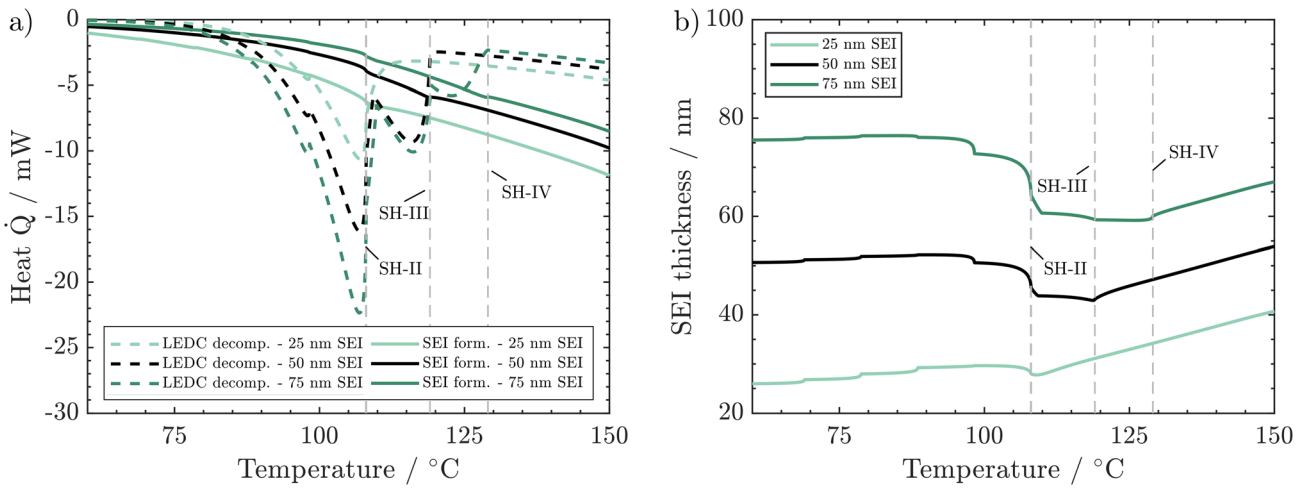

Fig. 6 Effect of SEI thickness on behaviour during the ARC test: (a) produced heat from LEDC decomposition and SEI formation. (b) Corresponding changes in SEI thickness. Conditions: 260 ppm water, 45 vol% LEDC.

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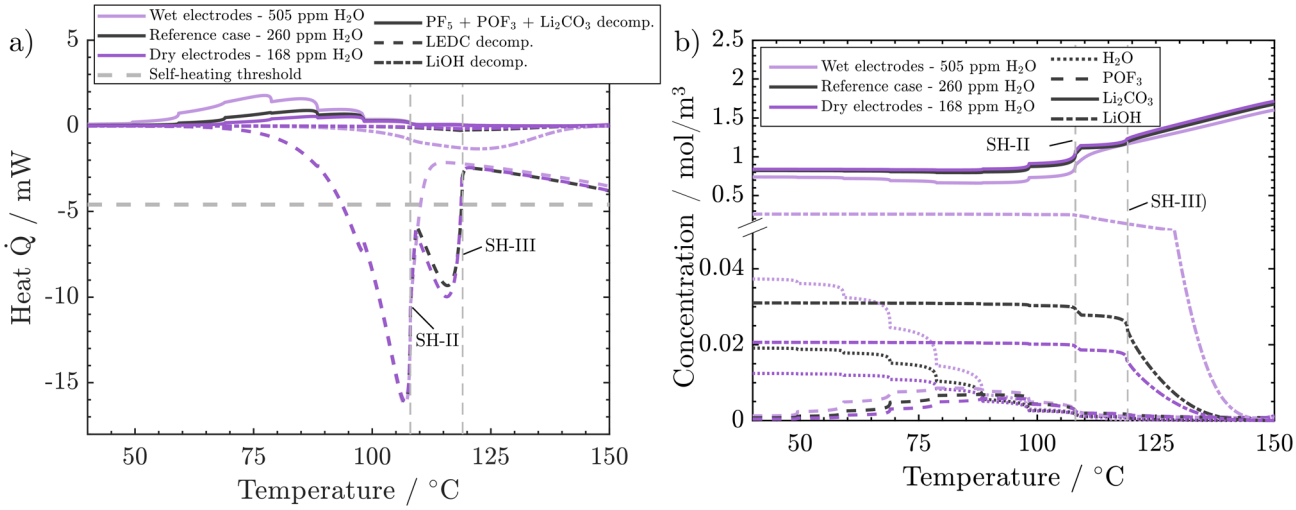

Fig. 7 Effect of H_2O impurities on behaviour during the ARC test: (a) combined endothermic heat of decomposition of PF_5, POF_3, and Li_2CO_3, *vs.* exothermic heat of LEDC and LiOH decomposition. (b) Concentrations of H_2O, POF_3, Li_2CO_3, and LiOH. Conditions: 45 vol% LEDC, 50 nm SEI.

the wet case the temperature gradient decreases, which is counterintuitive at first but is also connected to the higher LiOH content of 5 vol%. The significantly higher concentration of LiOH in the wet electrode case leads to notable LiOH decomposition rates and exothermic heat in the self-heating critical temperature frame of 100–130 °C. This leads to a transition into self-heating one temperature step earlier. The lower temperature in turn leads to a slower progression of all occurring reactions and, thus, a lower temperature gradient. This points to an important characteristic of an ARC measurement. As discrete temperature steps are used, more produced heat can either increase the temperature gradient (dry → medium dry), when self-heating starts during the same time step. Or it leads to a lower self-heating temperature, with a subsequent lower temperature gradient. Thus, it is important to discuss both characteristics together. In order to illustrate these impacts, the water variation case was simulated using a 2.5 °C temperature step instead of the 10 °C used before (see the ESI†). Even though the wet case also transitions into self-heating one step earlier, the gradients, 2.5210 °C h^{-1} (dry), 2.5459 °C h^{-1} (medium) and 2.6975 °C h^{-1} (wet), now better correspond to the produced heat as the temperatures are closer together.

In conclusion, H_2O impurities only play a significant role in battery safety when the electrodes have not been dried properly. The impact is negligible as soon as even a medium-intense drying procedure (260 ppm residual H_2O) is applied. Also, further drying (168 ppm residual H_2O) does not bring any noticeable benefit, because already at medium rates, LiOH decomposition rates are too low to give a significant heat contribution to tilt the balance towards sustained heating. As such, at least under the analysed circumstances and battery chemistry, extensive drying is unnecessary for a safer performance. This result aligns with the experimental findings of Huttner *et al.*,[32] who found a negative impact on performance metrics for intense drying procedures. Two more points should be accounted for in a holistic analysis of battery safety, which may be followed in further studies: the acids produced from H_2O, such as HPO_2F_2, are present in higher concentrations, *i.e.*, 0.175 mol m^{-3} in the wet case compared to 0.078 mol m^{-3} in the reference case. The acids were reported to lead to increased dissolution of transition metals from the cathode, which will then promote SEI decomposition.[57] Eventually, even though the energetic impact of these reactions is small, all are gassing reactions and will impact cell pressure.

3.3 Impact of variation in the SEI and H_2O content

So far, we have analysed how the thermal safety behaviour changes when varying a single variable, SEI thickness, composition or water content. In reality, multiple factors change due to ageing or different manufacturing processes. In the following, the impact of the cross-influence of the three variables is analysed and trends and generalisations are deduced.

Fig. 8a shows the temperature gradient in the region between self-heating on-set and thermal runaway, $\left(\frac{T_{TR} - T_{SH}}{t_{TR} - t_{SH}}\right)$, over the inverse self-heating temperature, $\left(\frac{1000}{T_{SH}}\right)$, for all 27 variations. Here, a high temperature gradient refers to a fast heating rate and, thus, a lower time to intervene. A low self-heating temperature indicates a lower resistance of the battery against thermal abuse. Thus, both characteristics can be considered indicators of battery safety and have therefore been chosen for this comparison. Cases that are located in the lower left corner of the figure can be considered rather safe, because it represents high self-heating temperatures and a low self-heating rate, whereas, the cases closer to the upper right show the opposite characteristics and, thus, can be considered rather critical. There are cases in all four quadrants of the figure, so no general correlation between the cases with fast self-heating and self-heating temperature can be found. The graph shows that the highly critical variations almost exclusively include high

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